SC-EADNet: A Self-Supervised Contrastive Efficient Asymmetric Dilated Network for Hyperspectral Image Classification

Mingzhen Zhu, Jiayuan Fan, *Member, IEEE*, Qihang Yang, and Tao Chen, *Member, IEEE*

***Abstract*— Unsupervised and semisupervised feature learning has recently emerged as an effective way to reduce the reliance on expensive data collection and annotation for hyperspectral image (HSI) analysis. Existing unsupervised and semisupervised convolutional neural network (CNN)-based HSI classification works still face two challenges: underutilization of pixel-wise multiscale contextual information for feature learning and expensive computational cost, for example, large floating-point operations per seconds (FLOPs), due to the lack of lightweight design. To utilize the unlabeled pixels in the HSIs more efficiently, we propose a self-supervised contrastive efficient asymmetric dilated network (SC-EADNet) for HSI classification. There are two novelties in the SC-EADNet. First, a self-supervised multiscale pixel-wise contextual feature learning model is proposed, which generates multiple patches around each hyperspectral pixel and develops a contrastive learning framework to learn from these patches for HSI classification. Second, a lightweight feature extraction network EADNet, composed of multiple plug-and-play efficient asymmetric dilated convolution (EADC) blocks, is designed and inserted into the contrastive learning framework. The EADC block adopts different dilation rates to capture the spatial information of objects with varying shapes and sizes. Compared with other unsupervised, semisupervised, and supervised learning methods, our SC-EADNet provides competitive classification performance on four hyperspectral datasets, including Indian Pines, Pavia University, Salinas, and Houston 2013, but few FLOPs and fast computational speed.**

***Index Terms*— Asymmetric dilated convolution, efficient asymmetric dilated network (EADNet), hyperspectral image (HSI) classification, self-supervised contrastive feature learning.**

I. Introduction

HYPERSPECTRAL image (HSI) classification is an important application in remote sensing, which is crucial for a wide range of thematic applications in modern society, such as ecological science, geological science, hydrological science, and precision agriculture. It aims at predicting the labels of the surface targets by using dense electromagnetic spectral information. The information is collected by high-resolution sampling in hundreds of observation frequency bands. Traditional methods mainly focus on spectral signatures extraction and developing different strategies, such as support vector machines (SVMs) [1], random forest (RF) [2], extreme learning machine (ELM) [3], and sparse representation [4], for HSI classification.

Due to the advancement of deep learning over the last few years, lots of HSI classification methods [5]–[7] based on deep convolutional neural networks (CNNs) have been developed and become mainstream, such as ResNet [8] in deep multiview learning (DMVL) [5], DenseNet [9] in multiscale densely-connected fusion networks [6], Res2Net [10], and nonlocal block [11] in spectral–spatial–semantic network with the multiway attention mechanism [7]. These methods typically contain two steps: 1) designing a robust feature extraction (FE) backbone to reduce the interference of spectral variations caused by sensor noise and enhance the discrimination of features between various classes and 2) designing a classification head to classify the extracted feature vectors. Compared with traditional handcrafted feature-based methods [1]–[4], CNN-based methods [5]–[7] can automatically generate features with richer representation and use millions of model parameters to fit more HSI surface targets better. Despite the good performance improvement on HSI classification, these methods still suffer from two problems.

First, many previous works [12]–[14] are based on supervised learning and require large amounts of labeled data. However, the collection and annotation of HSIs are expensive and time-consuming. Even though some recent works [5], [15], [16] start to exploit unsupervised or semisupervised methods for HSI classification, they mainly rely on a single pixel to extract spectral information or rely on multiple patches with fixed scales to extract the object semantic and spatial information. Therefore, such methods do not fully consider the pixel-wise contextual information around a target object pixel to discover the object semantic information and ignore the network design of variable and flexible receptive fields to extract multiscale spatial information from objects of varying shapes and sizes. Considering that the surface targets on the earth usually have different scales and shapes, such pixel-wise

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multiscale contextual information is crucial and should be considered for HSI classification.

Second, most previous HSI classification works [7], [17], [18] focus on how to design a deep and complex CNN model to achieve good performance. Little effort is devoted to lowering the computational complexity of the developed model, for example, reducing the floating-point operations per seconds (FLOPs) and parameter amount as well as increasing inference speed of the network. In fact, the HSI analysis has a high response time requirement when it is performed on power-constrained edge devices, including unmanned aerial vehicles, drones, and satellites equipped with various imaging sensors. Therefore, designing a lightweight HSI classification model with rapid inference speed and low memory cost should also be considered.

In view of these, we propose a self-supervised contrastive efficient asymmetric dilated network, namely SC-EADNet, for HSI classification. First, the original HSI image undergoes the principal component analysis (PCA) to generate a channel-compressed image with less redundant information. Then, multiple patches of different scales centering around a target pixel are sampled from the compressed image. Each pair of these patches is forwarded to the proposed self-supervised contrastive learning framework for learning features of different object classes. Finally, a linear classifier is concatenated and applied to these features to perform HSI image classification. A lightweight and efficient asymmetric dilated network (EADNet) is designed and inserted in the proposed contrastive learning framework. The EADNet consists of multiple plug-and-play efficient asymmetric dilated convolution (EADC) blocks, which use a parallel bottleneck convolution structure to extract rich spatial information of objects with different scales and shapes and is shown to achieve a good tradeoff between accuracy and the inference speed.

To summarize, our contributions are listed as follows.

1) We propose a self-supervised contextual feature learning framework at the pixel-wise multiscale level for HSI classification. The unlabeled data is used to learn the feature representation of HSIs by generating multiscale patches around each hyperspectral pixel. A contrastive learning technique is developed and integrated into the framework to learn from these patches for model optimization.
2) We develop a lightweight and EADNet, which is composed of multiple EADC blocks. The EADNet serves as the backbone of the whole framework and allows efficient FE that contains rich spatial information of objects with different shapes and sizes while keeping network parameter amount at a low level.
3) Thorough experiments are conducted on four hyperspectral datasets, including Indian Pines, Salinas, PaviaU, and Houston 2013. Our SC-EADNet shows competitive performance compared with other state-of-the-art supervised, semisupervised, and self-supervised methods, with few parameters and low computational cost.

This article is organized as follows. In Section II, we first review existing supervised, semisupervised, and self-supervised HSI classification works, and several typical lightweight CNN models. Section III gives an overview of the proposed SC-EADNet framework and its detailed components, including the proposed self-supervised contrastive learning technique, the developed EADNet structure, and the computational complexity analysis of the whole model. Experimental settings and results are given in Section IV. And lastly, a conclusion is made in Section V.

II. Related Work

A. Supervised and Semisupervised HSI Classification

Over the last few years, lots of supervised image classification methods [19]–[23] based on deep learning have been proposed and become the mainstream. Some recent works [6], [24], [25] exploit adjacent pixels as input, which simultaneously contain spectral and spatial information. In addition, some recent works [26], [27] also exploit semisupervised learning for HSI analysis, which combines active learning and pseudo-label method. All these supervised and semisupervised methods have verified that spatial information contributes to the performance improvement of CNN-based HSI classification methods. However, they still require large amounts of labeled samples and are not able to dig out good feature representations without labels.

B. Self-Supervised HSI Classification

Self-supervised learning [28]–[30] has achieved great success in computer vision field in recent years. Although self-supervised learning does not require handcrafted labels, it needs to design tasks to help the model learn some features in the data. The learned features are used in downstream tasks such as classification, detection, and segmentation. Self-supervised learning is roughly divided into two categories: generative learning and discriminative learning.

1) Generative Learning: Generative learning uses the self-similarity of data and constructs a model that can map original data to the specific feature. Such a feature, according to a certain distribution, can be used to reconstruct and represent the initial data. There are two famous generative learning models: Auto-encoder [31] and generative adversarial network (GAN) [32]. Inspired by Auto-encoder, [33]–[36] are based on the encoder–decoder paradigm, which first transforms the input HSI patch into a low-dimensional feature by a convolutional subnetwork (encoder) and then reconstructs the feature to initial data by a deconvolutional subnetwork (decoder). Inspired by GAN, [37]–[41] are based on the generator and discriminator paradigm. The generator generates samples that are not distinguishable by the discriminator, and the discriminator determines whether or not a sample is composed of real data.

2) Discriminative Learning: Discriminative learning focuses on modeling the difference between various samples but lacks the exact representative capability for each sample. There are two primary ways to implement discriminative learning: learning the model by solving a specific proxy task and contrastive learning. HSI feature learning network (HSINet) [42] uses the consistency among features at

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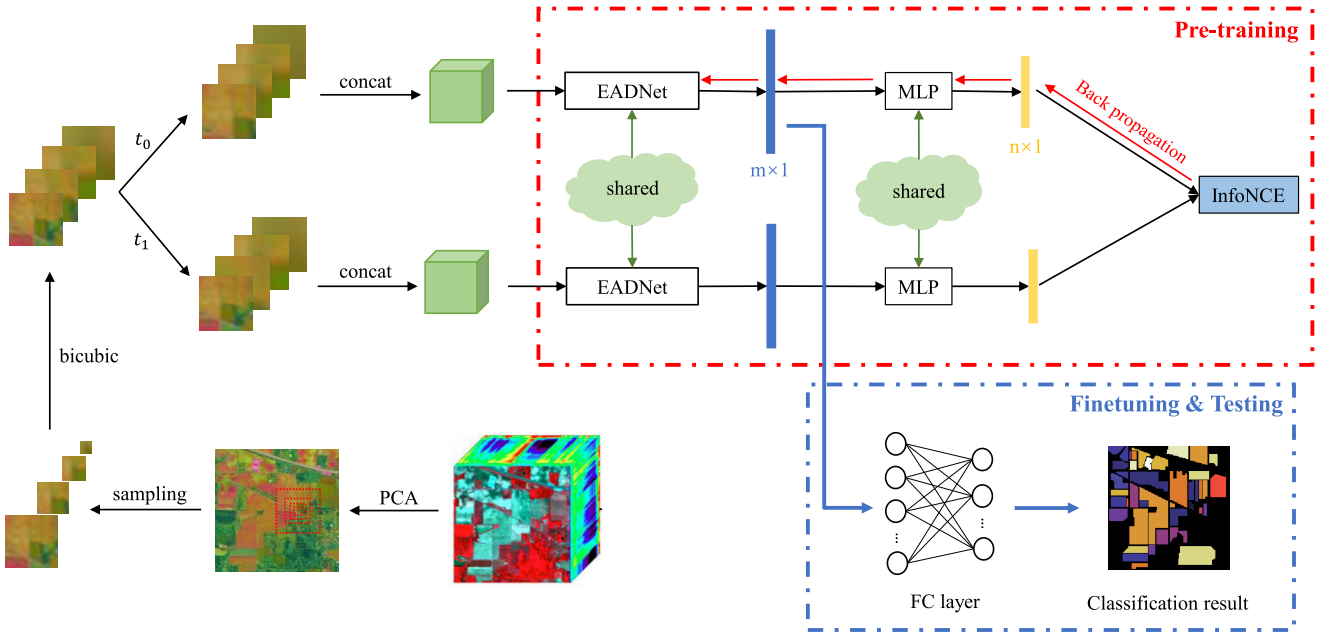


Fig. 1. Overview of the proposed SC-EADNet. t_0 and t_1 represent two branches of data augmentation used to generate different views for self-supervised contrastive learning. We use EADNet to extract representation vectors of length $m \times 1$ for each patch and use MLP to project the representation vectors to length $n \times 1$. InfoNCE is the optimization objective of the pretraining stage. The parameters of EADNet and MLP are shared between two branches. Classification is performed through an FC layer.

subpixel-level, superpixel-level, and pixel-level as constraints on learning a feature extractor. However, this way needs to manually design the proxy task, which relies heavily on the experience of experts. DMVL [5] uses the multiview, which are selected from different spectral bands of HSIs to generate positive samples. However, the multiview images generated from different spectral bands only have a color difference, which leads to the neural network cannot learn a good representation of HSIs. Inspired by SimCLR [29] and MoCo [28], in our proposed method, data augmentation is considered as an effective way to help contrastive learning, which is more robust and suitable for downstream tasks.

C. Real-Time Convolution Neural Network

Over the past few years, a couple of lightweight CNNs have been developed in semantic segmentation and object detection tasks. To simultaneously reduce computational cost and increase inference speed, previous works propose some efficient CNN architectures including reducing the times of downsampling [43], cascading multiscale feature map [44], [45], asymmetric convolution [46], depthwise separable convolution [47], and dense connection [48], [49]. However, the study on lightweight network design in the HSI classification task is still scarce. Inspired by U-shape network (UNet) [50], we use a skip connection to connect the detailed information in shallow layers and abstract semantic information in deep layers, which gets medium-sized feature maps with more details for HSI objects. Moreover, we have designed a lightweight EADNet [51] recently, which can extract the spatial information of objects with varying shapes and sizes and embed it into the self-supervised framework to ensure that our model can learn more representative features with a low computational cost.

III. Method

This section will introduce the proposed SC-EADNet in detail, including the proposed self-supervised contrastive feature learning framework and the developed lightweight FE backbone EADNet. As mentioned above, EADNet is a specially designed lightweight network, which has multiscale asymmetric receptive fields. The EADNet, used as the FE backbone, is further integrated into the self-supervised contrastive learning framework, SC-EADNet, to realize fast HSI classification. The following will give a detailed description.

A. Proposed SC-EADNet Overview

The overview of the proposed SC-EADNet is shown in Fig. 1. Redundant channels in HSIs may bring a high computational cost to the subsequent FE. Therefore, we first compress the original HSI by principal components analysis (PCA) to remove the redundant information. Then, in the pretraining stage, multiscale patches around the target object pixel are sampled from the compressed image to represent the contextual information around the target. A pair of patches denoting the same target object is forwarded to the subsequent contrastive learning framework, consisting of pairwise EADNet as FE backbone, followed by the multilayer perceptron (MLP) structure for feature projection. The InfoNCE [52] function is used as the optimization objective to pull the pair of patches belonging to the same target closer to each other and push the pair of patches belonging to different targets

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far away from each other. Fig. 1 shows that the proposed framework consists of two key modules in the pretraining stage: contrastive learning and EADNet. In the finetuning and testing stage, we apply a fully connected (FC) layer as a classifier after the pretrained backbone to extract features for HSI classification.

B. Self-Supervised Contrastive Feature Learning

The designed self-supervised contrastive feature learning framework contains three steps. We first use a series of data preprocessing operations for each hyperspectral pixel to extract the rich multiscale contextual information. Then, such information is forwarded to train the FE backbone EADNet by self-supervised contrastive learning. Note that the proposed self-supervised contrastive feature learning framework is not only applicable for the lightweight EADNet, but also suitable for other larger backbone networks, such as ResNet50. At last, we follow the linear evaluation protocol to evaluate the results. The detailed design of the feature learning framework is analyzed as follows.

1) Pixel-Wise Multiscale Contextual FE: As shown in Fig. 1, due to that the redundant channel information in HSIs may bring high computational cost, we first transform HSIs by PCA. Similar to [5], [6], [53], the first three principal components are taken to generate the original view for subsequent FE. Previous CNN-based HSI classification methods generally adopt fixed-size patches to extract spatial information, but such single-scale patches may not reflect the complex spatial structural information in HSIs. Inspired by MS-DenseNet [6], we use multiscale images as input data to enhance the multiscale contextual details. Centering around each hyperspectral pixel, we crop four different sizes of patches and set the patch sizes to be 3×3, 9×9, 15×15, and 31×31, respectively. Then we resize them to the exact size of 31×31 by bicubic interpolation. The patches of four scales cropped around the same pixel are the data example denoting the pixel's semantic category. A stochastic data augmentation module is then developed to transform a given data example into two correlated views, regarded as a positive pair. A positive pair represents the same object but in different views. Meanwhile, the views transformed from other data are considered as a negative pair. In this work, we apply one data augmentation: random cropping. At last, we concatenate the augmented contextual information from four scales of patches together and feed it into our FE backbone.

a) Data augmentation.: While data augmentation has been used in both supervised and unsupervised HSI feature learning, it has not been fully considered by the contrastive learning task in previous HSI classification works. Our work uses data augmentation to generate two correlated views of a target object. To the best of our knowledge, this is the first time to apply data augmentation in a systematic way for contrastive learning in the HSI analysis task. SC-EADNet learns a robust representation of the object by maximizing feature similarity between different augmented views of the same object.

We consider two types of augmentations here. One type involves the spatial transformation of data, such as crop, rotation, and flip. Another type involves appearance transformation, such as color distortion (including color dropping, brightness, contrast, saturation, hue), Gaussian blur, and random grayscale. The combination of different data augmentations plays an important role in learning good representations. Therefore, we study the performance of the proposed SC-EADNet by applying different augmentations individually or in pairs. As shown in Fig. 1, we use t_0 and t_1 two branches of data augmentations. In this experiment, we apply an individual or pairwise combination of data augmentations to two branches and evaluate the performance by K-nearest neighbors (KNN) classifier. We implement this experiment on the Indian Pines dataset. Here, 50% of data is for training, and the remaining 50% of data is for testing.

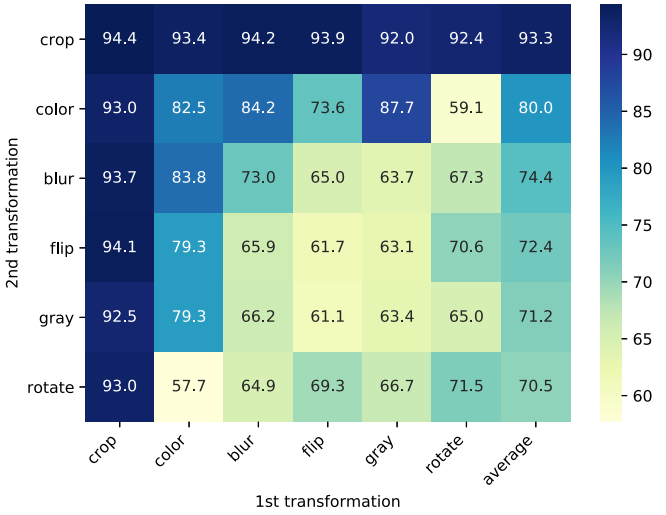


Fig. 2. Linear evaluation results on the Indian Pines dataset under individual or pairwise combination of data augmentation. The last column denotes the average value over each row. Best viewed in color.

Fig. 2 shows the linear evaluation results under individual or composition of transformations. We observe that one augmentation stands out: random crop. A simple but effective random crop operation can achieve the best performance. We believe the reason is the local feature consistency of the HSI. To be more specific, most pixels in a small HSI patch represent the same object class. When we randomly crop the patch, we can get contextual information with different spatial distributions but representing the same object, which is suitable for self-supervised contrastive learning. Besides, other data augmentation, such as color distortion and Gaussian blur, will destroy the spectral information of HSIs, which is essential for HSI classification. Therefore, a single random crop operation is adopted as our default data augmentation in the proposed self-supervised contrastive training framework.

2) Self-Supervised Contrastive Learning: We use the developed EADNet, detailed in Section III-C, as our base encoder to extract representation vectors from two augmented views. The representation vector's size is $m \times 1$. Then inspired by SimCLR [29], we use a simple MLP with one hidden layer as the projection head to further map the $m \times 1$ representation vector to a new $n \times 1$ space where the contrastive loss InfoNCE

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is computed. Here, following the setting in SimCLR [29] and DMVL [5], we set m to 2048 and n to 128.

The InfoNCE [52] loss is adopted as our contrastive loss function. For a positive pair of examples (i, j), the loss is defined as follows:

$$l_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\Sigma_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)} \quad (1)$$

where $\mathbb{1}_{[k \neq i]} \in \{0, 1\}$ is an indicator function which equals to 1 if $k \neq i$, and $\text{sim}(z_i, z_j) = z_i^T z_j / \|z_i\| \|z_j\|$ denotes the cosine similarity between two vectors z_i and z_j. τ denotes a temperature parameter, the default is set to 0.5. The total loss is computed across all positive pairs within a mini-batch.

3) Linear Evaluation Protocol: To evaluate the HSI classification performance based on the proposed self-supervised feature learning method, we follow the widely used linear evaluation protocol [5], [15], [35]. Once the self-supervised contrastive feature learning is completed, the backbone network can be an effective feature extractor. Therefore, we fix the parameters of the FE backbone and concatenate a classifier at the end of the backbone to classify the extracted feature vectors. We use an FC layer as a classifier to make the network easier to train and reduce the model complexity. The classifier is trained in a supervised manner with very few labeled samples.

C. EADNet Structure

The EADNet is the backbone feature extractor in the proposed framework, and it plays a crucial role in the whole HSI classification task. The design of EADNet pursuits strong FE ability and is extremely lightweight. The structure of EADNet is clear and simple without using any complex operation like attention mechanism [11], [54], or post-processing as illustrated in Fig. 3. The input of EADNet is concatenated augmented multiscale image patches, which contain both detailed and global information of objects.

As shown in Fig. 3, we employ two types of downsampling blocks, including ConcatConv Block, SeqDwConv Block, to extract features. The EADC block, inserted after the second and third ConcatConv blocks, is designed for efficient multiscale multishape FE corresponding to different receptive fields. n_0 and n_1 denote the repetitive number of EADC blocks in the EADNet. The feature map output from the last deep-layer EADC block and shallow-layer downsampling blocks, for example, the SeqDwConv block in the figure, are concatenated together and go through a 1×1 convolution to get the final feature map. Finally, the channel of the feature map is flattened to one dimension as the input of the MLP classifier. The EADNet has only three downsampling stages and obtains one-eighth origin image resolution like [55]. The detailed design of EADNet is analyzed as follows.

1) EADC Block: As the HSI often contains too many ground objects with arbitrary sizes and shapes, traditional convolution (e.g., regular 3×3 or 1×1 kernel) cannot sufficiently capture the features of these objects. Therefore, to solve the above problem, we propose the EADC block, a lightweight convolution unit with few parameters and FLOPs with asymmetric multiscale multishape receptive fields. The

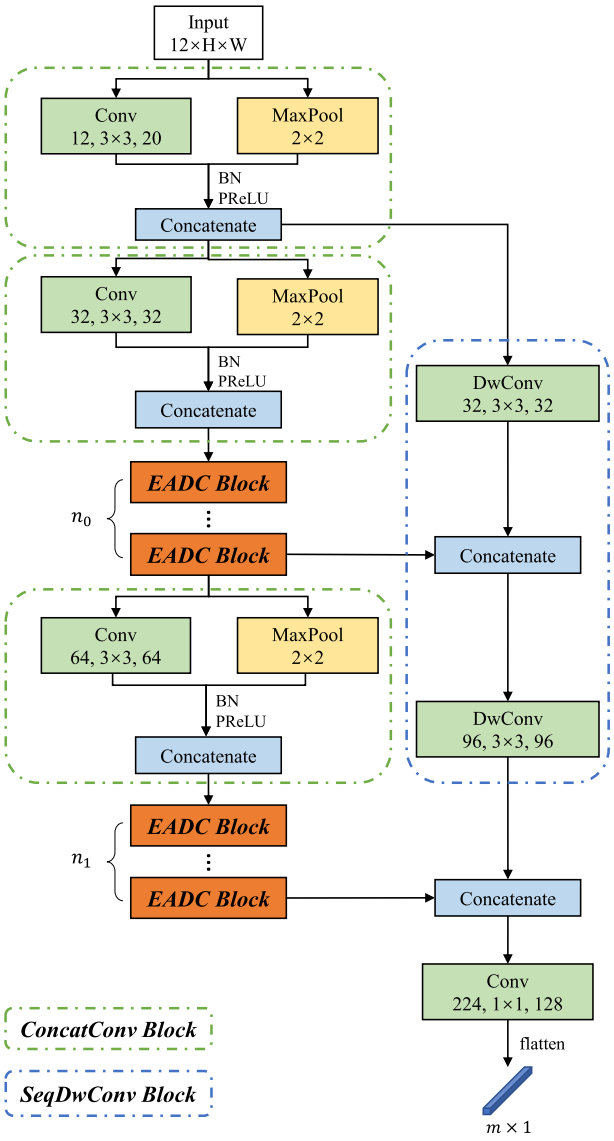


Fig. 3. Overview of the proposed EADNet. In the figure, DwConv means depthwise convolution, and n_i represents the repetitive number of EADC blocks. ConcatConv block is used for the backbone FE and downsampling. SeqDwConv block unifies the feature maps in shallow layers and concatenates them with the deep layers for FE. Best viewed in color.

EADC block consists of two parts: multiscale and multiratio receptive field convolution unit (MMRFC) and linear transform and fusion convolution unit (LTFC), as shown in Fig. 4. The dr in the figure denotes the base dilation rate of the EADC block in the current layer.

The MMRFC unit contains four convolution branches, where each branch is composed of three convolution operations: a 1×1 pointwise convolution, a 3×1 asymmetric (depthwise) convolution, and a 1×3 (depthwise) asymmetric convolution, following a parametric rectified linear unit (PReLU) [56]. In each branch, 1×1 convolution is designed to reduce the input dimension. The 3×1 and 1×3 asymmetric (depthwise) convolution with the same or different dilation rates are designed to achieve receptive fields of various shapes and sizes. In EADC block and EADNet, we adopt

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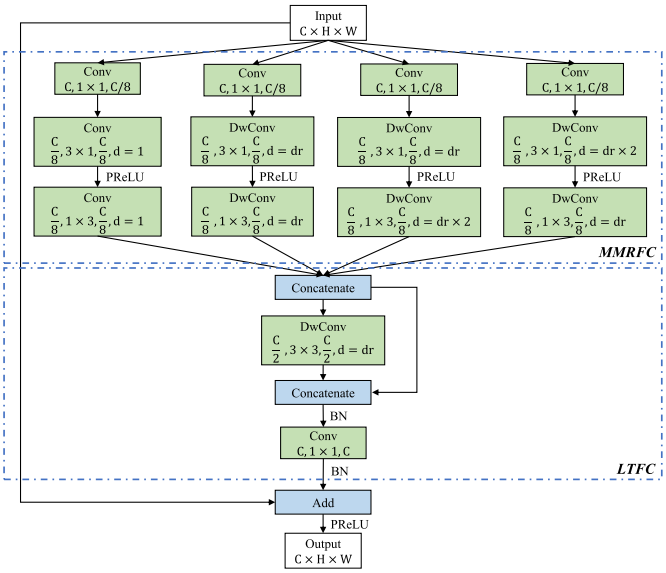


Fig. 4. Internal structure of the EADC block adopts a residual structure and contains two parts: the MMRFC unit and the LTFC unit. The shape of input feature maps for the EADC block is the same as the output, and dr denotes the base dilation rate of the EADC block in the current layer. The output of the LTFC unit is concatenated with the input of the EADC block and goes through PReLU [56] as the output of the EADC block.

PReLU [56] as an activation function, which has trainable parameters to adjust its slope and has a stronger ability to extract abstract information. It should be noted that the dilation rate in four branches is different for generating different scales of receptive fields, and we use asymmetric convolution in the first branch and use asymmetric depthwise convolution in the last three branches. The detailed design principles behind the MMRFC unit in the EADC block are given as follows.

a) Dilation rate versus filter weights.: The multiscale and multishape receptive fields are generated in the four branches of the MMRFC unit. Therefore, we need to set a suitable dilation rate to ensure that these asymmetric convolution filters produce the required receptive fields in the four branches. When the dilation rate gets larger, for example, beyond 15, the 1×3 or 3×1 filter may degenerate to a 1×1 filter since only the center weight is effective as analyzed in [57]. According to this, we adopt four concatenated dilation rate groups in four branches which are (1,1), (dr, dr), $(dr, dr \times 2)$, and $(dr \times 2, dr)$, respectively, where dr represents the base dilation rate of the EADC block in the current layer of EADNet and the value of dr usually increases gradually as the layer goes deeper. To ensure every dilation convolution kernel works, the maximum value of dr is set to 3, yielding a maximum dilation rate of six for each asymmetric convolution in all EADC blocks.

b) FE ability versus parameters.: As a part of the EADC block, the MMRFC unit makes an optimized tradeoff between the FE ability and parameter size. Generally, the spatial and channel correlations between adjacent pixels in an HSI image are stronger than those between nonadjacent pixels. Therefore, depthwise convolution that ignores between-channel correlations is suitable for extracting features from nonadjacent pixels. According to this, we adopt traditional asymmetric convolution (3×1, 1×3) to extract spatial and cross-channel features from adjacent pixels in the first branch, whose filter has a dilation rate of 1, indicating the neighboring pixels are used for FE. In the second, third, and fourth branches, depthwise asymmetric convolutions with lower computation costs are used, as more nonadjacent and distant pixels with weak channel correlations are used to extract features. To reduce the computation cost (FLOPs), the channel number of feature maps feeding to asymmetric convolution in each branch is compressed to one-eighth of the original via the 1×1 convolution.

After concatenating all feature maps from four branches in the MMRFC unit, the total feature channel number becomes one-half of the original input. To compensate for the compressed feature map information, we are inspired by GhostNet [58] and propose a simple yet effective LTFC unit. LTFC unit comprises one 3×3 depthwise convolution, a parallel skip connection, and a 1×1 pointwise convolution. LTFC unit is designed to double the concatenated input feature maps and fuse their channel information. The input and the output of the 1×1 pointwise convolution are batch normalized [59]. Through this, the 3×3 depthwise convolution generates lots of new information with little additional cost and finally goes through feature fusion by 1×1 pointwise convolution. This new information is essential to spatial information extraction.

We use EADC block with various base dilation rates to extract multiscale multishape features in different layers of EADNet as shown in Fig. 3. For n_0 EADC blocks, their base dilation rates dr are all set to 1, and for n_1 EADC blocks, their base dilation rates are gradually increased in a circulation manner. To balance the model complexity and the computational cost, we set n_0 and n_1 to be six and nine experimentally in this work. Among the n_1 EADC blocks, we set their base dilation rates like 1, 2, 3, and loop three times.

2) Downsampling Blocks in EADNet: In EADNet, we design two types of downsampling blocks in different layers. The principle of downsampling blocks is to retain more critical information we are concerned about in the current layer when downsampling. The two downsampling blocks are introduced as follows.

a) ConcatConv block.: This block is used for the backbone FE and downsampling, consisting of two parallel branches. One branch is a 3×3 convolution with the stride of 2 to extract information and downsampling. The other branch is a max-pooling operation to keep the original information and achieve downsampling. The output of two branches is concatenated following batch normalization [59] and PReLU [56], and the activated feature is fed into the next convolution block. Such a structure can keep downsampled details of previous feature input while producing new transformed features.

b) SeqDwConv block.: This block comprises two sequential 3×3 depthwise convolutions with the stride of 2, achieving fast feature transformation with $4 \times$ downsampling. It is used after the first ConcatConv block and the n_0 EADC block

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to generate the low-level feature output to compensate for the details miss in the deep-layer output. Batch normalization and activation function is not used in this block to preserve the details of the shallow-layer feature maps.

It is seen that in Fig. 3, the SeqDwConv block unifies different shallow layers' high-resolution feature maps, and the output of this block is concatenated with the final EADC block output via skip connection to compensate for the missing of detailed deep layer features.

D. Computation Complexity of EADNet

As CNN usually contains millions of parameters, the analysis of HSI consumes much memory and has a high computational cost. Hence, we present the computational complexity of the proposed method in this section. As the major component of the proposed approach is the EADNet, we calculate the parameter amount and FLOPs of the proposed EADC block. And the parameter amount and FLOPs of the EADNet and SC-EADNet are further analyzed in Section IV.

As described above, the EADC block follows a similar strategy as bottleneck [8]. It first compresses the number of input channels to one-eighth in each of the four branches by a 1×1 convolution. And then, it performs concatenated convolution in each branch for FE. Lastly, it combines them and restores the channel number to the original by the LTFC unit. In addition to the initial and final pointwise convolution of the EADC block, the spatial information is extracted through the (asymmetric) depthwise convolution, and cross-channel information is reserved in the first branch of EADC. These strategies can significantly reduce the number of FLOPs and parameter amount in the EADC block. In the following part, we will give the detailed formulas of the FLOPs and parameter size computation.

For the EADC block, its FLOPs are computed based on two parts: MMRFC unit and LTFC unit. Assuming the shape of the input feature is $C \times W \times H$, where C, W, and H denote the channel number, width, and height of the feature map, respectively, we denote by P_{Mi} $(i = 1, 2, 3, 4)$ and P_L as the convolution operation parameters in the four branches of MMRFC unit and LTFC unit, respectively. The P_{total} and F_{total} denote the convolution operation's parameter amount and FLOPs of the whole EADC block, respectively. They are computed as follows:

$$P_{Mi} = \begin{cases} (C+1) \times \dfrac{C}{8} + (3 \times \dfrac{C}{8} + 1) \times \dfrac{C}{8} \times 2, & i = 1 \\ (C+1) \times \dfrac{C}{8} + (3+1) \times \dfrac{C}{8} \times 2, & i = 2, 3, 4 \end{cases} \tag{2}$$

$$P_L = (3 \times 3 + 1) \times \frac{C}{2} + (C+1) \times C \tag{3}$$

$$P_{\text{total}} = \sum_{i=1}^{4} s P_{Mi} + P_L \tag{4}$$

$$F_{\text{total}} = P_{\text{total}} \times W \times H. \tag{5}$$

The maximum number of input channels for the EADC block is set at 128 in our work. Therefore, when C is equal to 128, ignoring a small number of parameters and FLOPs in PReLU [56], batch normalization [59], the EADC block has a parameter amount of 27 360 and FLOPs of $27\,360 \times W \times H$, which is one-fifth of that for a single common 3×3 convolution as shown in Table I. In EADNet, we employ a series of EADC blocks with variable base dilation rates (denoted as dr) in the network to extract features, making our network accurate, small, and fast.

TABLE I

PARAMETER AMOUNT AND FLOPS COMPARISON BETWEEN A SINGLE 3×3 CONVOLUTION AND EADC BLOCK

Input	Method	Parameter	FLOPs
$128 \times W \times H$	EADC block	27360	$27360 \times W \times H$
$128 \times W \times H$	3×3 convolution	147584	$147584 \times W \times H$

IV. EXPERIMENTS

In this section, we evaluate the performance of the proposed SC-EADNet for HSI classification. We start with the description of the datasets we used. Then we introduce the experimental setting and compare it with other state-of-the-art supervised, semisupervised, and unsupervised feature learning methods. Furthermore, we compare the training and FE time of SC-EADNet with other CNN-based methods. At last, we analyze the parameter settings of the proposed method.

A. Dataset Description

To demonstrate the effectiveness of the proposed SC-EADNet, four HSI datasets, including Indian Pines, Salinas, University of Pavia, and Houston 2013 datasets, are used to evaluate the performance. We will give a brief introduction of datasets as follows. The detailed information is shown in Tables II and III.

1) Indian Pines: Indian Pines dataset was gathered by Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor over the Indian Pines test site of North-Western Indiana in 1992. It consists of 145×145 pixels (21 025 samples in total) and 200 bands in the wavelength range 0.4–2.5 μm with a spatial resolution of 20 m. The dataset contains 16 classes and 10 249 labeled samples, including two-thirds agriculture and one-third forest or other natural perennial vegetation. The bands covering the water absorption region (104–108, 150–163, and 220) are removed, and 200 out of the 224 bands are used in our experiment.

2) Salinas: Salinas dataset was collected by the 224-band AVIRIS sensor over Salinas Valley, CA, USA. It consists of 512×217 pixels (111 104 samples in total) and 204 bands with a spatial resolution of 3.7 m. The dataset contains 16 classes and 54 129 labeled samples, including vegetables, bare soils, and vineyard fields. Similar to the Indian Pines dataset, 20 water absorption bands (108–112, 154–167, and 224) are out of 224 bands. Thus, 204 bands are used in our experiment.

3) University of Pavia: University of Pavia (PaviaU) dataset was acquired by the ROSIS-03 sensor over an urban area, northern Italy. It consists of 610×340 pixels (207 400 samples in total) and 103 bands with a spatial resolution of 1.3 m.

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TABLE II

LAND COVER TYPE, THE NUMBER OF LABELED TRAINING SAMPLES AND TESTING SAMPLES OF THE INDIAN PINES, SALINAS, AND UNIVERSITY OF PAVIA DATASET

	Indian Pines			Salinas			University of Pavia		
Class	Land Cover Type	Training	Testing	Land Cover Type	Training	Testing	Land Cover Type	Training	Testing
1	Alfalfa	5	46	Weeds-1	5	2009	Asphalt	5	6631
2	Corn-notill	5	1428	Weeds-2	5	3726	Meadows	5	18649
3	Corn-mintill	5	830	Fallow	5	1976	Gravel	5	2099
4	Corn	5	237	Fallow-plow	5	1394	Trees	5	3064
5	Grass-pasture	5	483	Fallow-smooth	5	2678	Metal sheets	5	1345
6	Grass-trees	5	730	Stubble	5	3959	Bare Soil	5	5029
7	Grass-pasture-mowed	5	28	Celery	5	3579	Bitumen	5	1330
8	Hay-windrowed	5	478	Grapes	5	11271	Bricks	5	3682
9	Oats	5	20	Soil	5	6203	Shadows	5	947
10	Soybean-notill	5	972	Corn	5	3278			
11	Soybean-mintill	5	2455	Lettuce-4wk	5	1068			
12	Soybean-clean	5	593	Lettuce-5wk	5	1927			
13	Wheat	5	205	Lettuce-6wk	5	916			
14	Woods	5	1265	Lettuce-7wk	5	1070			
15	Buildings	5	386	Vinyard-untrained	5	7268			
16	Stone	5	93	Vinyard-trellis	5	1807			
	Total	80	10249	Total	80	54129	Total	45	42776

TABLE III

LAND COVER TYPE, THE NUMBER OF LABELED TRAINING SAMPLES AND TESTING SAMPLES OF THE HOUSTON 2013 DATASET

	Houston University		
Class	Land Cover Type	Training	Testing
1	Healthy grass	5	1251
2	Stressed grass	5	1254
3	Synthetic grass	5	697
4	Trees	5	1244
5	Soil	5	1242
6	Water	5	325
7	Residential	5	1268
8	Commercial	5	1244
9	Road	5	1252
10	Highway	5	1227
11	Railway	5	1235
12	Parking Lot 1	5	1233
13	Parking Lot 2	5	469
14	Tennis Court	5	428
15	Running Track	5	660
	Total	75	15029

The dataset contains nine classes and 42 776 labeled pixels, including asphalt, bricks, meadows, trees, and so on, 12 noisy bands are removed, and 103 out of the 115 bands are used in our experiment.

4) Houston 2013: Houston 2013 dataset was gathered by ITRES Research Ltd., CASI-1500 sensor over the Houston University campus and its neighboring area. It consists of 349 $\times$ 1905 pixels (664 845 samples in total) and 144 bands with a spatial resolution of 2.5 m. The dataset contains 15 classes and 15 029 labeled pixels, including grass, trees, soil, water, and so on.

B. Comparison Results With State-of-the-Art Methods

In this section, we compare the proposed SC-EADNet with seven state-of-the-art methods. To demonstrate the proposed method can significantly improve the classification accuracy in the case of small samples, we follow the experimental setting in DMVL [5]. In the feature learning procedure, 50% of the unlabeled samples are used as the training data, and the remaining 50% of the samples are used as the testing data. In the finetuning procedure, five labeled samples per class are randomly selected as the training samples for the supervised classifier. The details of the number of training and testing samples are shown in Table II.

The proposed SC-EADNet is designed, trained, and tested using the PyTorch framework. The widely used Adam [60] optimizer is adopted to optimize the proposed SC-EADNet. In the self-supervised contrastive pretraining procedure, the above-mentioned InfoNCE [52] loss is used as a loss function. And the batch size is set to be 64. The learning rate and weight decay are set to be 1e-3 and 1e-6, respectively. The number of epochs is set to be 50. In the finetuning procedure, we fix the parameters of EADNet and connect an FC layer at the bottom of EADNet to classify the extracted features. To finetune the classifier, the widely used cross-entropy loss is used as a loss function. And the batch size is set to be 10. The learning rate and weight decay are set to be 1e-3 and 1e-6, respectively. The number of epochs is also set to be 50. In addition, following the experiment setting in [5], [35], and [61], a fivefold cross-validation strategy is used for evaluation.

Compared methods include the traditional FE method extended morphological profile (EMP) followed by SVM [62], supervised method joint collaborative representation (JCR) [63], semisupervised method transductive support vector machine (TSVM) [64], supervised method MS-DenseNet [6], and self-supervised methods 3-D-convolutional autoencoder (CAE) [35], GAN [65], and DMVL [5]. Note that we only use five labeled samples for training for all methods. The details of compared methods are listed as follows.

1) EMP + SVM [62] is a traditional spatial FE method that could improve the classification accuracy. As for EMP, two commonly used morphological filters based on square structure elements (opening and closing) are used to construct the morphological attribute profiles. The radius of structuring elements is set to be 1, 3, 5, 7, and 9, respectively.
2) TSVM [64] is a semisupervised method that could use unlabeled samples to improve classification accuracy. All unlabeled samples are used for training.

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TABLE IV

COMPARISON RESULTS ON THE INDIAN PINES DATASET. FIVE LABELED SAMPLES ARE RANDOMLY SELECTED FOR TRAINING. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class No.	EMP+SVM	TSVM	JCR	3D-CAE	GAN	MS-DenseNet	ResNet50	DMVL	EADNet	SC-ResNet50	SC-EADNet
1	32.23	86.96	97.83	86.96	52.38	100.0	100.0	86.96	100.0	100.0	100.0
2	55.14	36.20	35.15	32.91	48.51	49.50	64.07	67.51	54.62	61.06	55.04
3	37.86	42.65	56.02	23.13	45.94	68.79	42.53	72.53	52.40	78.31	66.74
4	40.51	53.59	45.57	45.15	39.61	92.40	78.90	99.58	100.0	97.04	87.34
5	33.95	78.88	69.98	51.14	46.44	55.48	79.91	68.12	53.62	83.43	81.15
6	78.41	70.82	85.34	75.62	93.01	72.73	73.01	76.16	50.27	87.53	93.15
7	29.17	92.86	100.0	100.0	39.71	100.0	100.0	100.0	100.0	100.0	100.0
8	95.18	85.15	90.38	84.94	97.36	100.0	90.79	100.0	99.58	98.32	94.56
9	21.28	90.00	100.0	100.0	25.32	100.0	75.00	100.0	85.00	100.0	100.0
10	41.34	41.56	70.27	54.42	47.67	50.30	57.51	83.64	56.79	85.69	91.97
11	71.86	55.15	59.92	53.24	68.69	55.35	58.65	71.41	70.10	64.27	88.39
12	25.20	50.08	43.68	34.06	35.18	54.13	52.27	49.07	46.71	86.00	80.43
13	88.72	96.10	100.0	85.85	78.13	100.0	85.85	98.05	100.0	100.0	100.0
14	75.28	82.77	91.46	71.30	89.90	83.71	78.33	99.05	71.46	88.06	87.74
15	66.05	35.32	59.07	23.58	57.42	96.63	81.08	87.56	95.33	98.70	93.52
16	100.0	93.55	100.0	88.17	100.0	100.0	90.32	100.0	96.77	100.0	100.0
OA(%)	56.80	57.65	64.95	52.21	60.24	66.02	66.05	78.01	65.97	78.76	**82.69**
AA(%)	55.76	68.22	75.29	63.15	60.33	79.94	75.51	84.98	77.04	**89.28**	88.75
Kappa	55.12	52.24	60.66	46.20	55.38	61.84	62.05	75.31	61.94	76.31	**80.37**

TABLE V

COMPARISON RESULTS ON THE SALINAS DATASET. FIVE LABELED SAMPLES ARE RANDOMLY SELECTED FOR TRAINING. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class No.	EMP+SVM	TSVM	JCR	3D-CAE	GAN	MS-DenseNet	ResNet50	DMVL	EADNet	SC-ResNet50	SC-EADNet
1	95.36	97.31	97.86	92.19	99.65	100.0	100.0	95.92	87.92	100.0	100.0
2	98.85	96.32	98.36	99.68	72.84	98.84	83.41	100.0	71.35	99.81	97.39
3	81.09	91.24	99.80	58.96	77.88	75.25	100.0	100.0	97.97	85.12	92.44
4	97.74	99.57	97.42	96.41	99.07	99.28	99.92	100.0	100.0	75.32	94.67
5	96.37	94.47	92.91	89.66	96.38	98.39	98.80	94.73	94.31	99.92	100.0
6	100.0	99.77	97.58	98.89	98.94	99.92	98.76	90.96	87.88	99.79	100.0
7	94.52	98.02	96.28	97.46	92.71	95.55	82.31	98.60	81.78	99.83	99.58
8	81.25	59.94	82.60	70.46	49.71	63.89	78.22	91.47	70.88	88.61	85.49
9	99.06	95.31	99.90	94.66	94.57	98.59	97.82	99.92	91.72	99.32	99.83
10	86.86	83.59	80.57	69.89	89.08	87.52	88.77	80.63	87.13	100.0	99.72
11	68.27	97.19	98.41	91.10	96.72	95.31	96.62	99.25	95.57	100.0	99.71
12	88.64	100.0	100.0	97.98	100.0	97.82	94.96	94.91	78.72	99.68	99.63
13	60.11	97.82	91.92	98.03	97.82	100.0	99.01	72.49	96.26	99.78	100.0
14	95.20	97.20	98.50	81.59	95.98	98.41	75.70	98.97	92.86	97.94	99.34
15	55.55	79.27	33.17	64.93	85.43	63.26	60.18	95.56	54.6	84.61	89.27
16	88.47	86.05	98.62	79.80	70.56	99.94	100.0	100.0	100.0	99.22	96.28
OA(%)	83.47	85.63	84.93	52.21	81.67	85.05	85.98	94.60	80.23	94.18	**94.73**
AA(%)	86.71	92.07	91.49	63.15	88.58	92.00	90.90	94.59	86.93	95.56	**97.08**
Kappa	81.68	84.09	83.14	46.20	79.75	83.39	84.45	94.00	77.03	93.52	**94.14**

TABLE VI

COMPARISON RESULTS ON THE PAVIAU DATASET. FIVE LABELED SAMPLES ARE RANDOMLY SELECTED FOR TRAINING. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class No.	EMP+SVM	TSVM	JCR	3D-CAE	GAN	MS-DenseNet	ResNet50	DMVL	EADNet	SC-ResNet50	SC-EADNet
1	72.16	69.78	74.08	63.26	60.91	89.39	87.93	57.80	89.24	87.63	85.32
2	86.13	66.25	60.28	90.02	77.47	81.52	88.59	98.32	93.16	84.40	95.46
3	80.75	65.03	75.08	60.60	69.56	43.64	53.88	84.37	69.31	99.95	61.55
4	98.40	62.14	81.63	80.00	92.46	97.51	83.19	56.01	81.49	97.25	95.20
5	99.26	100.0	100.0	99.85	98.36	99.70	84.90	100.0	100.0	100.0	99.33
6	16.88	84.97	70.77	12.79	48.98	50.87	57.56	100.0	52.35	92.68	45.43
7	96.54	98.27	88.42	83.23	95.94	90.11	98.37	99.85	61.42	100.0	90.07
8	8.23	80.07	84.06	43.10	55.43	92.41	97.71	97.23	63.36	97.17	80.52
9	99.79	54.17	97.57	97.04	99.89	99.36	77.19	27.35	84.16	99.89	98.41
OA(%)	70.77	71.62	70.90	70.85	72.06	80.59	83.47	86.96	82.21	**89.98**	85.06
AA(%)	73.13	75.63	81.32	69.99	77.67	82.72	81.03	80.10	77.17	**95.44**	83.48
Kappa	61.35	64.71	63.94	61.01	64.03	74.56	78.18	82.46	76.25	**87.08**	79.80

3) JCR [63] is a representation-based classification method. In this method, neighbors near the test pixel are simultaneously represented via linear combinations of available training samples.

4) 3-D-CAE [35] is a self-supervised spatial–spectral feature learning method based on a 3-D CAE. The parameters are set the same as the article. The patch size is set to be 28×28.

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TABLE VII

COMPARISON RESULTS ON THE HOUSTON 2013 DATASET. FIVE LABELED SAMPLES ARE RANDOMLY SELECTED FOR TRAINING. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class No.	EMP+SVM	TSVM	JCR	3D-CAE	GAN	MS-DenseNet	ResNet50	DMVL	EADNet	SC-ResNet50	SC-EADNet
1	85.29	95.20	71.30	83.37	84.01	80.05	49.49	50.52	64.62	81.44	81.80
2	94.90	85.49	85.25	81.98	66.99	80.33	46.01	68.74	38.85	94.56	67.53
3	99.57	99.00	92.97	99.43	41.75	100.0	93.96	86.94	93.33	99.87	99.87
4	93.17	91.40	89.31	69.69	87.78	90.98	48.57	63.42	35.44	88.60	94.46
5	85.43	97.26	97.18	97.83	87.04	69.10	66.40	87.84	65.17	95.99	98.84
6	76.62	92.62	85.85	73.85	40.31	88.20	66.66	75.38	90.26	99.41	87.02
7	58.99	43.61	69.09	75.39	54.97	97.69	69.24	92.35	81.84	84.55	78.79
8	25.32	34.41	24.12	60.21	32.23	47.71	60.70	64.31	41.06	46.30	36.48
9	71.09	53.75	61.02	66.29	16.29	71.04	45.10	81.87	34.74	81.33	71.62
10	29.58	57.78	56.89	44.09	60.39	40.09	46.76	100.0	49.22	95.22	91.08
11	70.69	71.74	60.32	78.79	49.31	71.83	67.17	88.34	63.02	82.88	93.35
12	18.33	51.26	78.26	58.56	51.26	55.63	33.86	68.45	31.84	35.54	61.72
13	34.54	17.70	30.06	74.41	36.03	89.71	75.94	95.74	86.86	94.62	92.08
14	99.07	96.50	84.58	99.77	89.72	96.88	99.80	100.0	80.11	100.0	98.83
15	98.79	97.88	99.85	82.12	73.33	78.19	72.93	82.73	68.17	98.24	100.0
OA(%)	67.02	70.68	71.30	74.36	58.60	74.00	58.59	78.55	56.43	**82.14**	80.89
AA(%)	69.42	72.37	72.40	76.39	58.09	77.16	62.84	80.44	61.64	**85.24**	83.56
Kappa	64.42	68.33	69.04	72.32	55.25	71.96	55.48	76.81	53.29	**80.73**	79.39

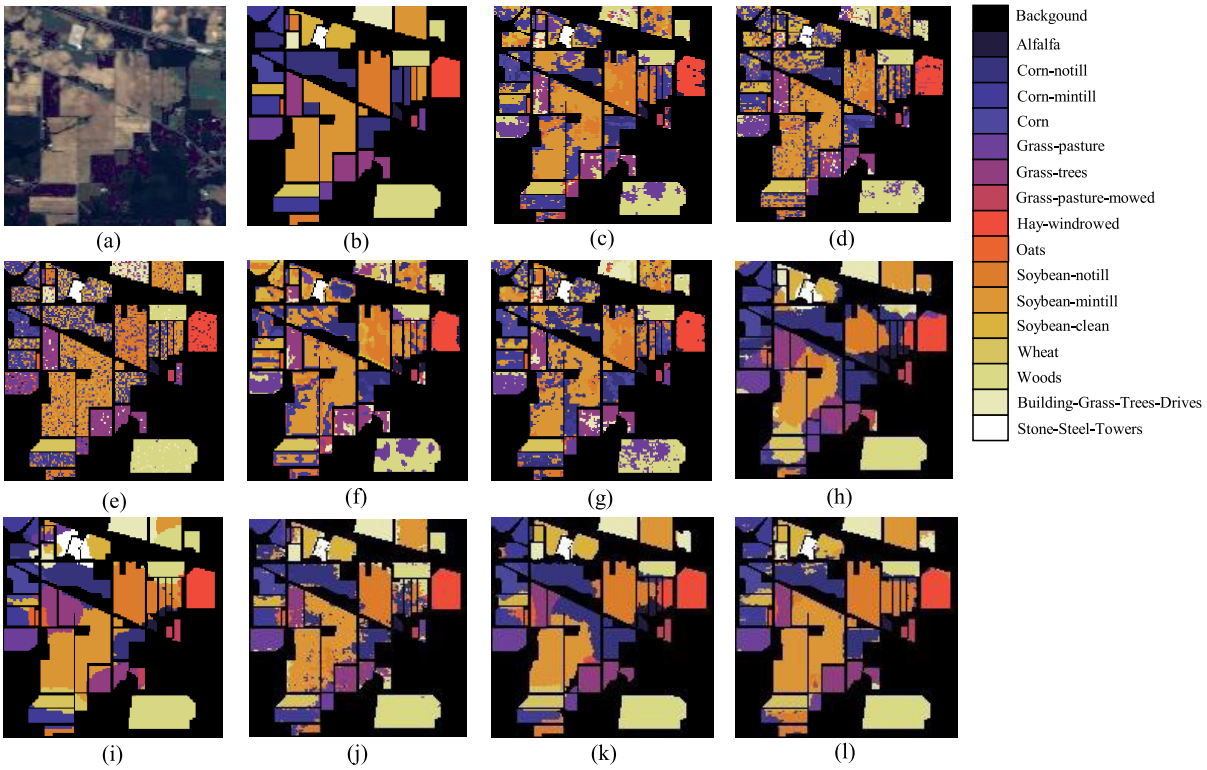


Fig. 5. Classification maps of different methods on the Indian Pines dataset. (a) Pseudo-color image. (b) Ground-truth map. (c) EMP + SVM (OA: 56.80%). (d) TSVM (OA: 57.65%). (e) JCR (OA: 64.95%). (f) 3-D-CAE (OA: 52.21%). (g) GAN (OA: 60.24%). (h) ResNet50 (OA: 66.05%). (i) DMVL (OA: 78.01%). (j) EADNet (OA: 65.97%). (k) SC-ResNet50 (OA: 78.76%). (l) SC-EADNet (82.69%).

5) GAN [65] is a self-supervised feature learning method based on 2-D GAN. PCA is used to reduce the dimension of HSIs to three dimensions. The patch size is set to be 28×28.
6) MS-DenseNet [6] is a supervised method based on densely connected convolution network [9], which also uses multiple scale patches as input. PCA is used to reduce the dimension of HSIs to three. The patch sizes are set to be 3×3, 9×9, 15×15, and 31×31.
7) DMVL [5] is a self-supervised contrastive learning method. Two views of an HSI scene are constructed by applying PCA to different bands. Then, ResNet50 [8] is adopted as a backbone network to extract spatial features, which are fed into an SVM classifier to get the classification results. The patch size is set to be 28×28.
8) To further evaluate the effectiveness of our proposed backbone network and self-supervised contrastive learning technique, we first train a supervised EADNet

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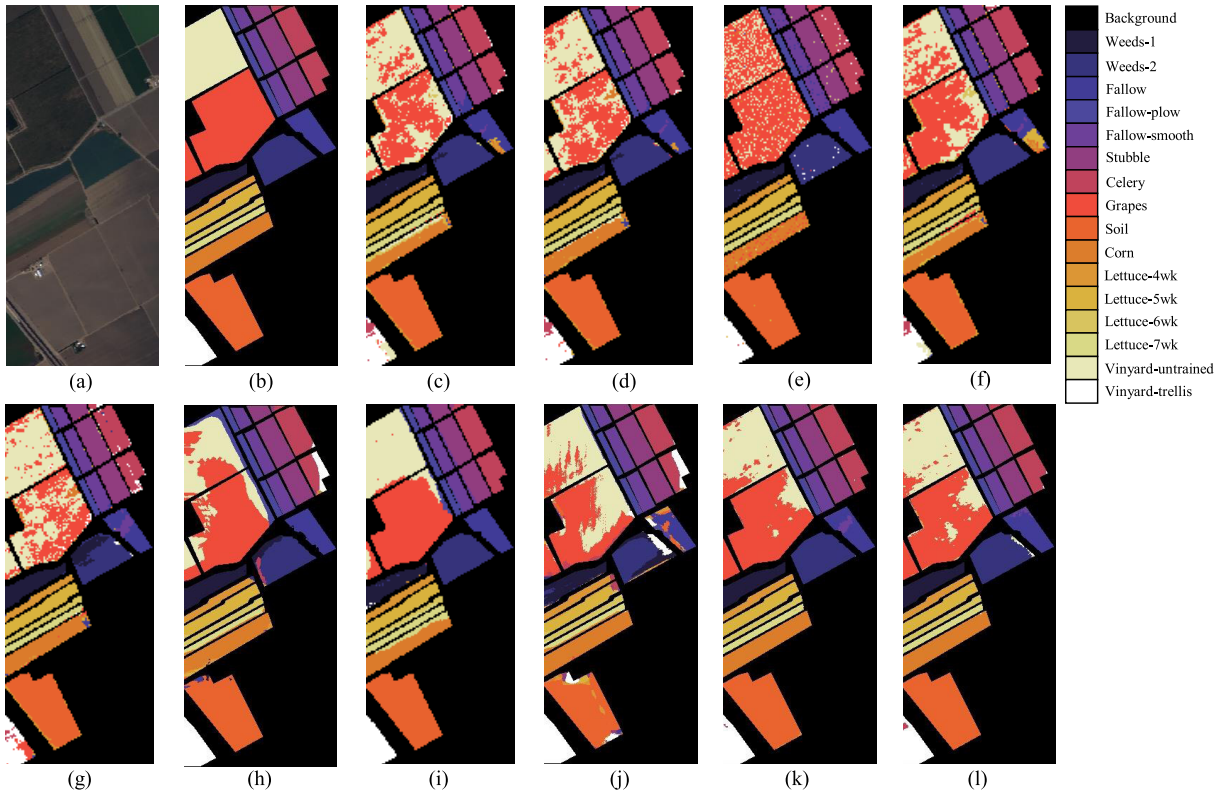


Fig. 6. Classification maps of different methods on the Salinas dataset. (a) Pseudo-color image. (b) Ground-truth map. (c) EMP + SVM (OA: 83.47%). (d) TSVM (OA: 85.63%). (e) JCR (OA: 84.93%). (f) 3-D-CAE (OA: 52.21%). (g) GAN (OA: 81.67%). (h) ResNet50 (OA: 85.98%). (i) DMVL (OA: 94.60%). (j) EADNet (OA: 80.23%). (k) SC-ResNet50 (OA: 94.18%). (l) SC-EADNet (94.73%).

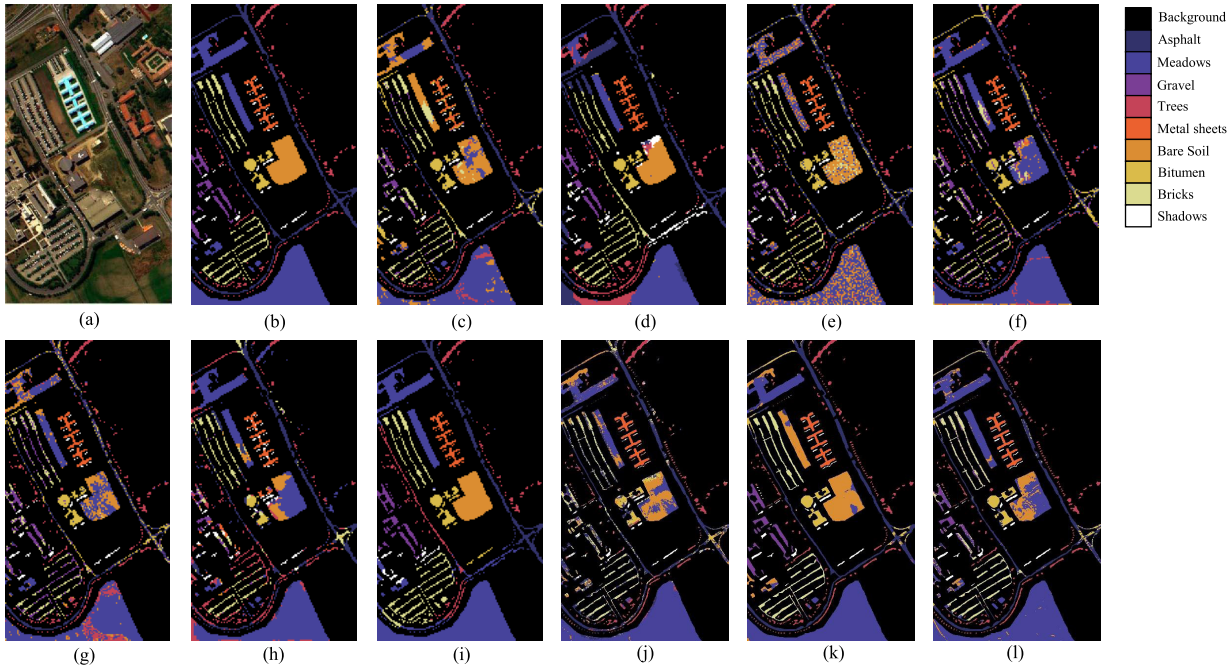


Fig. 7. Classification maps of different methods on the PaviaU dataset. (a) Pseudo-color image. (b) Ground-truth map. (c) EMP + SVM (OA: 70.77%). (d) TSVM (OA: 71.62%). (e) JCR (OA: 70.90%). (f) 3-D-CAE (OA: 70.85%). (g) GAN (OA: 72.06%). (h) ResNet50 (OA: 83.47%). (i) DMVL (OA: 86.96%). (j) EADNet (OA: 82.21%). (k) SC-ResNet50 (OA: 89.98%). (l) SC-EADNet (OA: 85.06%).

classifier and a supervised ResNet50 classifier with five labeled training samples per class. The input of ResNet50 and EADNet are both a single patch with the size of 31×31. The learning rate, batch size, and epochs are set to 1e-3, 10, and 100, respectively. Then, we change the backbone with the ResNet50 classifier

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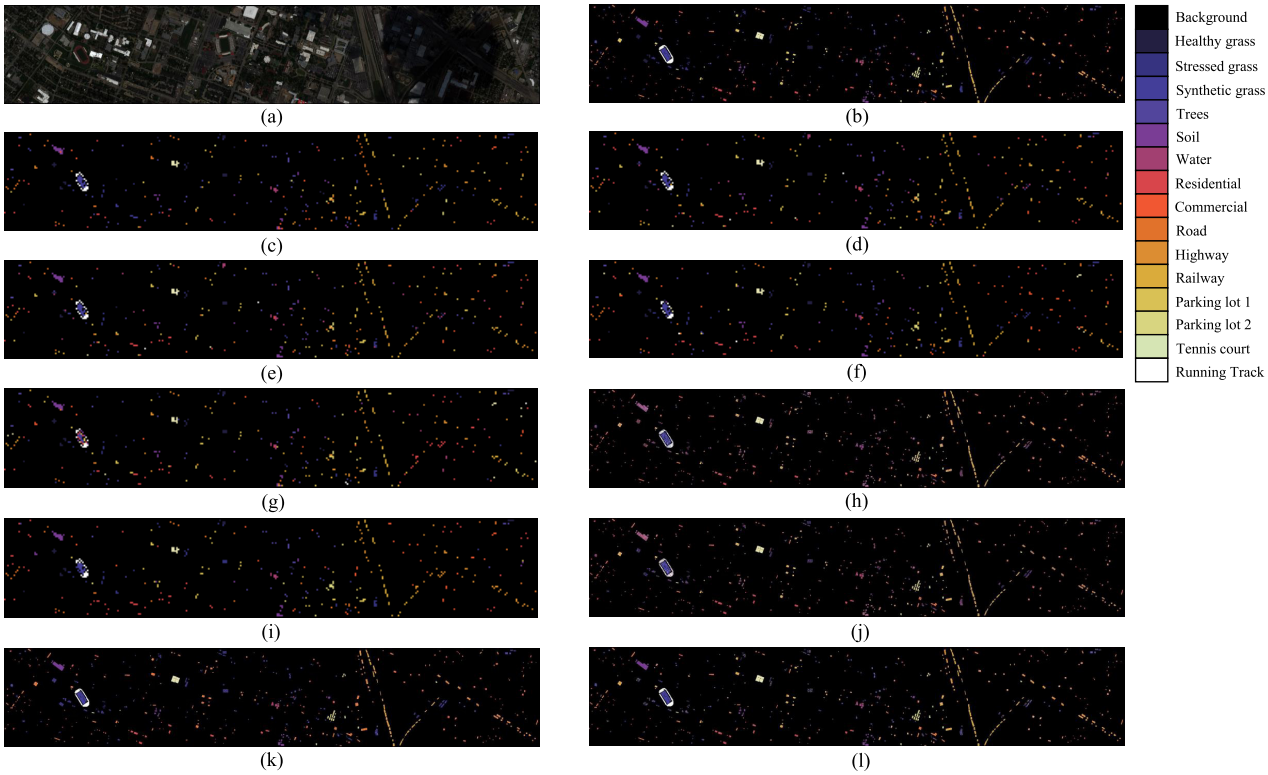


Fig. 8. Classification maps of different methods on the Houston 2013 dataset. (a) Pseudo-color image. (b) Ground-truth map. (c) EMP + SVM (OA: 67.02%). (d) TSVM (OA: 70.68%). (e) JCR (OA: 71.30%). (f) 3-D-CAE (OA: 74.36%). (g) GAN (OA: 58.60%). (h) ResNet50 (OA: 58.59%). (i) DMVL (OA: 78.55%). (j) EADNet (OA: 56.43%). (k) SC-ResNet50 (OA: 82.14%). (l) SC-EADNet (OA: 80.89%).

to observe the performance of the SC-ResNet50. The input of SC-EADNet and SC-ResNet50 are both four multiscale patches with the size of 3×3, 9×9, 15×15, and 31×31. And the setting of the learning rate, batch size, and the number of epochs are the same as SC-EADNet.

The class-specific accuracy, overall accuracy (OA), average accuracy (AA), and Kappa of different methods on the Indian Pines, Salinas, PaviaU, and Houston 2013 datasets are listed in Tables IV–VII, respectively. And the classification map of different methods on the four datasets are shown in Figs. 5–8, respectively. Some data in the table is quoted from [5].

Compared with other methods, our proposed SC-EADNet and its variant SC-ResNet50, by changing the backbone, consistently achieves the highest OA, AA, and kappa on three datasets. For example, in Table IV on the Indian Pines dataset, the OA of SC-EADNet (82.69%) yields over 20% higher than EMP + SVM (56.80%) and TSVM (57.65%) and over 15% higher than MS-DenseNet (66.02%). Compared with other self-supervised methods, our proposed SC-EADNet also provides competitive results. Especially on the Indian Pines dataset, SC-EADNet achieves the best OA of 82.69%, which is higher than the counterpart of 3-D-CAE (52.21%), GAN (60.24%), and DMVL (78.76%). On the Salinas dataset, SC-EADNet also achieves the best OA of 94.73%, which is higher than the counterpart of 3-D-CAE (52.21%), GAN (81.67%), and DMVL (94.60%). On the PaviaU dataset, SC-EADNet gets the OA of 85.06%, which is much higher than the counterpart of 3-D-CAE (70.85%) and GAN (72.06%). On the Houston 2013 dataset, SC-EADNet gets the OA of 82.14%, which is much higher than the counterpart of 3-D-CAE (74.36%), GAN (58.60%), and DMVL (78.55%).

Note that the number of unlabeled training samples on the PaviaU and Houston 2013 dataset is 103 700 and 332 422 (half of the total samples), which is much more than the other two datasets. Due to the proposed SC-EADNet is in an extremely lightweight design, its representative capability for such a large amount of training samples may not be as good as ResNet50-based models. Therefore, on the PaviaU dataset, the OA of SC-EADNet (85.06%) is slightly inferior to that of DMVL (86.96%), which uses ResNet50 as the backbone network on the dataset. However, on the Houston 2013 dataset, the OA of SC-EADNet (82.14%) is still better than that of DMVL (78.55%), but the number of parameters of ResNet50 is 60 times larger than that of EADNet. When we replace our backbone network with ResNet50 for a fair comparison, SC-ResNet50 could achieve better OA than SC-EADNet on the PaviaU (89.98%) and Houston 2013 (82.14%). And they are higher than ResNet50-based models such as DMVL on these two datasets (86.96% and 78.55%). Hence, the proposed SC-EADNet is competitive by achieving the best tradeoff between accuracy and inference speed. And its variant SC-ResNet50 could achieve a better result on large-scale datasets than other methods, which shows the effectiveness of the proposed self-supervised contrastive learning technique.

Furthermore, we can observe that the proposed EADNet classifier achieves the OA of 65.97%, 80.23%, 82.21%, and 56.43%, respectively, on the Indian Pines, Salinas, PaviaU, and

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Houston 2013 datasets. As its counterpart, the OA of ResNet50 is 66.05% (+0.07%), 85.98% (+5.75%), 83.47% (+1.26%), and 58.59% (+2.16%), respectively, which are better than EADNet. However, the number of the parameters and FLOPs of EADNet is much smaller than that of ResNet50, as shown in Table IX. It also shows that our proposed FE strategy, which extracts rich object spatial information of different shapes and sizes, is still efficient. Remarkably, compared with EADNet, SC-EADNet has great performance improvement after the self-supervised pretraining. On three datasets, the gaps are +16.72%, +14.50%, +2.85%, and +24.46%, respectively, where the gaps on the other datasets are larger than that on the PaviaU. The reason is that the objects on the Indian Pines, Salinas, and Houston 2013 datasets are spatially close regions and visually homogeneous, presenting stronger local feature consistency. Therefore, the proposed self-supervised contrastive learning technique brings great performance improvement on the two datasets. On the other hand, the objects on the PaviaU dataset are spatially sparse. However, the proposed self-supervised contrastive learning technique still provides about 3% increase in performance. It demonstrates that our self-supervised contrastive training technique can also perform well on the large-scale and locally sparse hyperspectral dataset.

As mentioned before, the proposed self-supervised contrastive training framework is flexible by replacing the backbone with different networks to adapt to varying scales of datasets. Thus, we replace our backbone as ResNet50, the same as DMVL, and show its results by SC-ResNet50. Note that DMVL generates different views by selecting different bands of HSI, while our proposed method generates different views by randomly cropping from HSI. SC-ResNet50 provides competitive results compared with DMVL. On the Indian Pines, Salinas, PaviaU, and Houston 2013 datasets, the OA of DMVL are 78.01%, 94.60%, 86.96%, and 78.55%, respectively. And the OA of SC-ResNet50 are 78.76% (+0.75%), 94.18% (−0.42%), 89.98% (+3.02%), and 82.14% (+3.59%). It demonstrates that our proposed self-supervised learning technique is flexible to adapt to datasets of various scales and present competitive performance improvement compared with models (DMVL) using the same backbones.

The above-mentioned experimental results have proved the effectiveness of the proposed self-supervised contrastive pretraining technique and the FE ability of the proposed EADNet. Primarily, SC-EADNet performs well in the case of a small number of labeled samples (only five per class).

C. Computational Complexity Analyses

This section compares the training time, FE time, parameter amount, and FLOPs of the proposed method with other CNN-based methods. Note that all experiments in this work are carried out using a PC equipped with Intel Core i7-9700K CPU and a single GPU of NVIDIA GeForce GTX 2080Ti.

1) Training and FE Time Analysis: The whole training time and FE time of Ms-DenseNet [6], 3-D-CAE [35], DMVL [5], and the proposed SC-EADNet are reported in Table VIII. Due to the lightweight design of the proposed EADNet, SC-EADNet achieves the fastest inference speed. On the Indian Pines, Salinas, and PaviaU datasets, the FE time of SC-EADNet is 4.6, 14.4, and 10.7 s, respectively, which is much faster than that of MS-DenseNet, 3-D-CAE, and DMVL. The training procedure of SC-EADNet takes more time than MS-DenseNet and 3-D-CAE because all unlabeled samples are used for self-supervised contrastive learning. However, the training procedure of SC-EADNet is still faster than DMVL on the Indian Pines and Salinas datasets. Considering the classification accuracy and inference speed, our proposed SC-EADNet is competitive.

TABLE VIII

EXECUTION TIMES OF TRAINING AND FE PROCEDURES IN THE INDIAN PINES, SALINAS, AND PAVIAU DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Indian Pines				
	MS-DenseNet	3D-CAE	DMVL	SC-EADNet
Training (min)	23.3	**19.2**	141.2	44.9
FE (s)	9.1	5.2	27.3	**4.6**
FE/pixel (μs)	894.7	509.3	2664.6	**453.7**
Salinas				
	MS-DenseNet	3D-CAE	DMVL	SC-EADNet
Training (min)	22.3	**19.3**	243.1	236.2
FE (s)	44.3	26.4	77.7	**14.4**
FE/pixel (μs)	819.7	487.7	1436.2	**267.5**
PaviaU				
	MS-DenseNet	3D-CAE	DMVL	SC-EADNet
Training (min)	22.3	**19.4**	197.7	440.0
FE (s)	36.0	32.0	40.8	**10.7**
FE/pixel (μs)	842.9	749.0	954.2	**250.3**

TABLE IX

PARAMETER AMOUNT AND FLOPS COMPARISON BETWEEN MS-DENSENET, RESNET-50, AND SC-EADNET. FC DENOTES THE FULLY CONNECTED LAYER. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Input	Model	Parameter (M)	FLOPs (M)
$4 \times 3 \times 31 \times 31$	MS-DenseNet	4.47	712.19
$3 \times 31 \times 31$	ResNet50	23.48	85.90
$4 \times 3 \times 31 \times 31$	EADNet	**0.39**	**9.25**
$4 \times 3 \times 31 \times 31$	EADNet w/ FC	0.42	9.30
$4 \times 3 \times 31 \times 31$	SC-EADNet	1.45	20.60

2) Parameter Amount and FLOPs Analysis: Compared with other typical backbone networks, such as MS-DenseNet [6] and ResNet50 [8] in DMVL [5], the proposed EADNet saves a lot of parameters and FLOPs due to the lightweight EADC block. A detailed comparison is given in Table IX. Our FE backbone EADNet has just 0.39 M parameters and 9.4 M FLOPs when the input size is $4 \times 3 \times 31 \times 31$, which is much lighter and faster than others. And for rigorousness, we report the parameter amount and FLOPs of our proposed EADNet with FC layer used in the finetuning and testing stage and the self-supervised contrastive learning framework SC-EADNet used in the pretraining stage. Table IX demonstrates that our FE backbone is much more lightweight than MS-DenseNet [6] and ResNet50 [8].

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TABLE X

RESULTS OF EADNET BY VARYING VALUES OF n_0, n_1 AND THE BASE DILATION RATE dr IN EADNET. "$*$" IN THE TABLE MEANS CIRCULATION OPERATION, FOR EXAMPLE, $(1, 2, 3) * 2$ MEANS THAT THERE ARE SIX EADC BLOCKS AND THEIR BASE DILATION RATES ARE 1, 2, 3, 1, 2, AND 3, RESPECTIVELY. EXPERIMENTAL RESULTS ARE REPORTED ON THE INDIAN PINES DATASET

n_0	n_1	dr of n_0	dr of n_1	FLOPs(M)	Parameter(M)	OA(%)
4	6	1*4	(1, 2, 3)*2	6.95	0.32	65.16
4	9	1*4	(1, 2, 3)*3	8.38	0.41	65.46
6	8	1*6	(1, 2, 3, 4)*2	8.97	0.39	64.84
6	9	1*6	(1, 2, 3)*3	9.45	0.42	**65.74**
6	9	(1, 2, 3)*2	(1, 2, 3)*3	9.45	0.42	64.78
6	12	1*6	(1, 2, 3, 4)*3	10.88	0.51	64.59
8	12	1*8	(1, 2, 3)*4	11.95	0.53	64.41

D. Ablation Study of EADNet

This section analyzes in detail the effect of each component on our proposed EADNet step by step. We train a supervised EADNet classifier with five labeled training samples per class to evaluate the performance. The results are reported on the Indian Pines dataset.

1) Ablation for EADC Block Number: Our network includes $n_0 + n_1$ EADC blocks, and different values of n_0 and n_1 will impact the performances of EADNet. The results by changing values of n_i and base dilation rate are shown in Table X. It can be seen that the best performance is achieved when $n_1 = 6$ and $n_2 = 9$. Smaller values of n_i cause the EADNet to only capture a limited receptive field size in the image, leading to poor performance on large size targets. On the other hand, larger values of n_i will increase network parameters and computation cost and make the network heavier and easy to overfitting.

2) Ablation for MMRFC: MMRFC captures the variable shape and size of receptive fields in the EADC block. To verify the effect of MMRFC in the EADC block, we replace MMRFC with a single branch of one 1×1 convolution followed by a 3×3 depthwise convolution, with the same base dr as MMRFC and keep other settings unchanged. The new results are shown in Table XI. It can be seen that the proposed MMRFC brings a 2.1% performance gain over EADNet without MMRFC.

3) Ablation for LTFC: LTFC performs a linear transformation on feature maps to recover more information. To verify the effect of LTFC in the EADC block, we replace LTFC with a simple 1×1 convolution to restore the number of output channels with other settings unchanged. The new results are shown in Table XI. It can be seen that the proposed LTFC brings 2.6% performance gain over EADNet without LTFC.

4) Ablation for EADC Block: As a key fundamental block in EADNet, the EADC block plays an important role in extracting objects of different shapes and sizes. To evaluate the effect of the EADC block, we replace the EADC block with a single branch of one 1×1 pointwise convolution followed by a sequential 3×3 depthwise convolution and 1×1 pointwise convolution. The comparison results are shown in Table XI. It can be seen that the proposed EADC block brings 3.6% performance gain over EADNet without EADC block.

5) Ablation for SeqDwConv Block: SeqDwConv block unifies and concatenates the features in the shallow and deep layers, effectively combining the image details and abstract information before prediction. We remove the SeqDwConv Block and skip connection of EADNet and then compare it with the original EADNet. The results are shown in Table XI. It can be seen that the SeqDwConv Block and DwConv Block can improve the segmentation results further by 1.1% over the EADNet without these two blocks.

TABLE XI

FLOPS, PARAMETER AMOUNT, AND OA COMPARISON BETWEEN ORIGINAL EADNET AND ITS VARIANTS BY CHANGING MMRFC, LTFC, EADC BLOCK, AND SKIP CONNECTION STRUCTURE. EXPERIMENTAL RESULTS ARE REPORTED ON THE INDIAN PINES DATASET

Method	FLOPs(M)	Parameter(M)	OA(%)
EADNet	9.45	0.42	**65.74**
EADNet w/o MMRFC	8.34	0.38	63.69
EADNet w/o LTFC	6.60	0.31	63.10
EADNet w/o EADC	6.28	0.29	62.16
EADNet w/o SeqDwConv	6.40	0.29	64.63

E. Parameter Analyses

In this section, we analyze the effect of single-scale patch sizes, the number of multiscale patches, and different numbers of training samples on the performance of our proposed method. We evaluate the performance by training the SC-EADNet with the same experimental setting as Section IV-B. The results are reported on the Indian Pines dataset.

1) Effect of Different Patch Sizes: In this section, we first evaluate the performance of the proposed SC-EADNet with different patch sizes. As shown in Fig. 9, the input patch size of the SC-EADNet is varied from 3×3 to 31×31. The best OA is 75.45% when the patch size is 15×15. As described in [6], a small patch can well capture the detailed structures, and a large patch can better represent the global distribution. However, too large patches containing multiple semantic classes will lead to a decline in performance due to the interclass interference in the patch. In order to take both detailed information and large contextual information into

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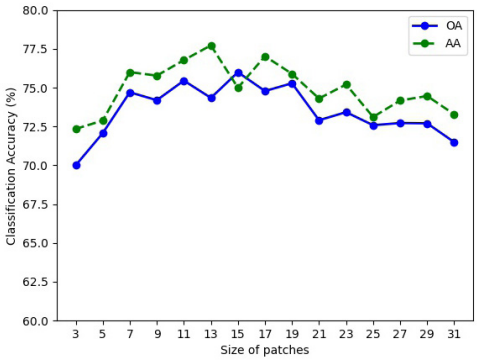


Fig. 9. Classification results of different patch sizes on the Indian Pine dataset.

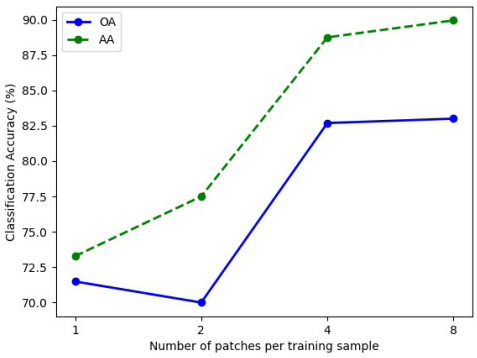


Fig. 10. Classification results of different number of input patches on the Indian Pine dataset.

consideration, we try to use multiscale patches as the input of SC-EADNet.

2) Effect of the Number of Multiscale Patches: This section evaluates the performance of pixel-wise multiscale contextual FE with different numbers of multiscale patches. To better extract multiscale contextual information, we use multiple patches with different scales as input of the SC-EADNet. We consider the number of input patches as 1, 2, 4, and 8. Referring to the previous works [5], [6], [35], to alleviate the interclass interference in a patch, our patch size is not larger than 31×31 pixels. To take both detailed information and large contextual information into consideration, we will select both small and large sizes of patches empirically in one patch group. The details of patch size selection are as follows. For one patch, 31×31 is selected. For two patches, 3×3 and 31×31 are selected. For four patches, 3×3, 9×9, 15×15, and 31×31 are selected. For eight patches, 3×3, 7×7, 11×11, 15×15, 19×19, 23×23, 27×27, and 31×31 are selected. The classification results on the Indian pines dataset are shown in Fig. 10. It can be observed that the proposed SC-EADNet achieves the best performance when the number of input patches is 8. Compared with the results of the four patches input, it improves the OA by 1.17% but also produces twice as many FLOPs. To balance the inference speed and the performance, we choose four patches in the proposed method.

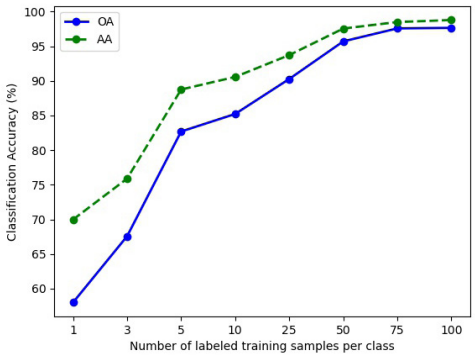


Fig. 11. Classification results of different number of labeled training samples on the Indian Pine dataset.

3) Effect of Different Number of Labeled Training Samples: This section further evaluates the effect of varying the number of training samples on classification performance. As the contrastive learning technique is in a self-supervised manner, the backbone of SC-EADNet can be trained without using labeled samples. Then we fix the parameters of the FE backbone and finetune a classifier at the end of the FE network with only a few labeled samples. The results mentioned above have shown that SC-EADNet performs well on an extremely small number of labeled training data. We vary the number of training samples for the classifier from 1 to 100. The classification results on the Indian Pines dataset are shown in Fig. 11.

It is observed that the classification accuracy increases steadily when more training samples are used to train the classifier. When the number of training samples per class is only 1, the OA of our proposed SC-EADNet is 58.01%. When the number of training samples per class reaches 100, the OA reaches 97.65%. This well validates that the proposed SC-EADNet can learn effective feature representation in a self-supervised manner.

V. Conclusion

In this work, we propose an SC-EADNet for HSI classification. The proposed network is pretrained under a self-supervised contrastive learning manner without using any labeled sample and finetuned with only a small number of labeled samples. We generate different views by randomly cropping on the small patches of HSIs. Due to the local feature consistency of HSIs, SC-EADNet can learn a good object representation by maximizing feature similarity between differently augmented views of the same object. To the best of our knowledge, this is the first work to incorporate data augmentation in a systematic way into the contrastive learning task in the HSI analysis field. On the other hand, the proposed FE backbone EADNet can capture the spatial information of remote-sensing objects with varying shapes and sizes. Because of its extremely lightweight design, it can achieve very few FLOPs and very fast inference speed. Experiments on four datasets demonstrate the FE ability of the proposed EADNet, and the corresponding HSI classification ability of the proposed SC-EADNet.

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